FBISE

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

COMPUTER SCIENCE

Past Paper 2022 • Paper Code: 2SA-I 2211 — D(6111)

Retyped faithfully from the original board paper — same questions, no changes.

Note: COMPUTER SCIENCE SSC-II. SECTION – A (Marks 12). Time allowed: 15 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 6111. Fill the relevant bubble against each question:

Q1. Which symbol is used for decision in a flow chart? (Note: four flow chart symbols are shown as options in the source.)

(A) (pentagon-like flow chart symbol)
(B) (rectangle)
(C) (diamond)
(D) (rounded rectangle)

کون سی علامت flowchart میں decision کے لیے استعمال ہوتی ہے؟

Q2. How many bytes are used for long int data type?

(A) 2
(B) 4
(C) 8
(D) 10

long int ڈیٹا ٹائپ کے لیے کتنے bytes استعمال ہوتی ہیں؟

Q3. Which operator is an arithmetic operator?

(A) + =
(B) %
(C) &&
(D) =

درج شدہ میں کون سی arithmetic operator کی علامت ہے؟

Q4. Which operator has lowest precedence?

(A) !=
(B) <=
(C) =
(D) %

کس علامت (Operator) کی precedence سب سے کم ہے؟

Q5. What is the output of following code segment after execution when a=1 and b=2? if (a+b++<4) printf("%d",b); else printf("%d",a);

(A) 1
(B) 2
(C) 3
(D) 4

درج شدہ Program code segment کا output کیا ہو گا؟

Q6. What is the output of following code segment after execution when a=10, b=−20: if (a>0&&b<0) printf("%d",a+b); else printf("%d",a−b);

(A) −10
(B) 10
(C) 20
(D) 30

درج شدہ Program code segment کا output کیا ہو گا؟

Q7. Which of the following is valid data type that can be used as variable/expression in switch statement?

(A) float
(B) double
(C) int
(D) unsigned

درج شدہ میں سے کون سی ڈیٹا ٹائپ switch statement میں variable/expression کے لیے مناسب ہے؟

Q8. Which loop structure is most suitable when number of iterations/repetitions are known in advance?

(A) while loop
(B) for loop
(C) do while loop
(D) nested loop

درج شدہ میں سے کون سا loop structure زیادہ مناسب ہے جب loop کے repetitions پہلے سے معلوم ہوں؟

Q9. Which of the following is unary gate?

(A) NOT
(B) AND
(C) XOR
(D) XNOR

درج شدہ میں کون سا unary gate ہے؟

Q10. Which combination of logic gates is used to make a NAND gate?

(A) AND & OR
(B) OR & NOT
(C) AND & NOT
(D) NOR & NOT

درج شدہ میں سے کون سا طالب NAND gate بنائے گا؟

Q11. How many heading tags are there in HTML language?

(A) 4 (B) 5

(C) 6 (D) 7

HTML Language میں کل کتنے Heading tags ہیں؟

Q12. Which heading tag pair is used to display largest size heading in an HTML web page?

(A) <h5></h5> (B) <h3></h3>

(C) <h4></h4> (D) <h6></h6>

درج شدہ میں سے کون سا heading tag pair سب سے بڑی heading دکھانے کے لیے HTML web page میں استعمال ہوتا ہے؟

> Note: —2SA-I 2211-6111 (HA)—. Page 1 of 2. Page 2 of 2.

> Note: COMPUTER SCIENCE SSC–II. Time allowed: 2:45 Hours. Total Marks Sections B and C: 43. NOTE: Answer any nine parts from Section 'B' and any two questions from Section 'C' on the separately provided answer book. Use supplementary answer sheet i.e. Sheet-B if required. Write your answers neatly and legibly. (The source also carries the full Urdu translation of this paper.)

Q. SECTION – B (Marks 27)

Q. 2 Answer any NINE parts from the following. All parts carry equal marks. (9 x 3 = 27)

(i) Write about the criteria for selecting the best solution of a problem.

(ii) What is meant by the following terms? (a) Algorithm (b) Efficiency of an Algorithm (c) Candid solutions

(iii) What is meant by the following terms? (a) Character set (b) Reserved words (c) Comments

(iv) Differentiate between constant and variable with one example of each.

(v) Differentiate between getch() and getche() functions by writing their syntax, purpose and giving an example of each.

(vi) What is format specifier? Differentiate between %f and %e format specifiers.

(vii) What is an arithmetic operator? Differentiate between simple and compound assignment operators with an example of each.

(viii) What is meant by the term nested selection structure. When do we need it?

(ix) What is loop? Write syntax of for loop and give an example.

(x) Write the following program code segment using while loop structure. void main(void) { int x, sum=0; for (x=1;x<=99;x=x+2) sum=sum+x; printf("Sum of odd numbers from 1 to 99 is %d",sum); }

(xi) Write the following program code segment using switch statement: if (a+b == 1 || a+b == 2 || a+b == 3) printf("you are in first 3 positions"); else printf("you are not in first 3 positions");

(xii) What is meant by the following terms? (a) World Wide Web (WWW) (b) Website (c) Uniform Resource Locator (URL)

(xiii) Write purpose of following HTML tags. (a) Font Size (b) Subscript (c) Strike-out

Q. SECTION – C (Marks 16)

Note: Attempt any TWO questions. (2 x 8 = 16)

Q. 3 a. Write syntax and explain purpose of switch and else-if statements. (04) b. Write/ convert following program code using if-else statement. void main (void) { char ch; printf("Enter any uppercase or lowercase letter from A to Z"); scanf("%c", &ch) { case 'a': case 'A': case 'e': case 'E': case 'i': case 'I': case 'o': case 'O': case 'u': case 'U': printf("You typed a vowel"); break; default: printf("You typed a consonant"); } }

Q. 4 a. What is meant by Truth Table? Draw/create a Truth Table for the Boolean function, F = x'z + y z' + x y z (04) b. What is Karnaugh Map? Simplify the given Boolean function using Karnaugh Map. F = A'B'C + A'BC + AB'C' + AB'C + ABC + ABC' (04) (Note: complement bars as in source.)

Q. 5 a. Describe the following websites. i. News ii. Information iii. Entertainment iv. Business (2+2+2+2)

> Note: —2SA-I 2211 (HA)—